

Executive Directive: End-to-end architect, implement, validate, and deploy the dual-domain **HealthRisk AI** platform and the gamified quarterly simulation engine **HealthRisk Lab**. This operational task list governs code deliverables, machine learning performance baselines, quantitative actuarial engines, governance protocols, and final submission compliance.

1. SYSTEM ARCHITECTURE DIRECTIVES

Core Mandates

Dual-Domain Integration

Translates clinical & epidemiological streams (MIMIC-IV, openFDA, WHO GH0, ClinicalTrials.gov) into financial risk models across Health Insurance, Hospital Credit, Pharma Portfolios, and ESG.

Gamified Simulation Engine

HealthRisk Lab: 40-quarter (10-year) turn-based simulation with 4 game modes, AI opponent benchmark, multi-shock scenario generators, and a 1000-point scoring system.

2. PHASE 1 – ENVIRONMENT & PIPELINE FOUNDATIONS (DAYS 1–4)

Infrastructure & Data

TASK ID	COMPONENT MODULE	IMPLEMENTATION SPECIFICATION	TARGET BENCHMARK	STATUS
INF-01	Repository & MLOps	Setup modular directory structure, <code>docker-compose.yml</code> , DVC data pipelines, and MLflow experiment logging tracking.	Docker build error-free	[] Pending
INF-02	CI/CD Pipeline	Configure <code>.github/workflows/ci-cd.yml</code> running ruff code formatting, mypy static typing, and pytest suite.	Green status on main	[] Pending
DAT-01	Data Ingestion APIs	Automated ingestion scripts in <code>data/acquisition/</code> fetching openFDA FAERS, ClinicalTrials.gov, WHO GH0, CDC WONDER.	Automated pipeline output	[] Pending
DAT-02	Feature Engineering	Build ICD-10 hierarchy maps, CMS-HCC risk adjusters, lab trajectory slopes (HbA1c/eGFR), and DRG/CMI financial indices in <code>data/features/</code> .	Modular transformations	[] Pending

3. AUTONOMOUS STATE MACHINE RULES

PROGRESS.md Protocol

- ☐ **State Tracking:** Initialize persistent `PROGRESS.md` at root on session launch. Mark task `[IN_PROGRESS]` prior to coding.
- ☐ **Validation Loop:** Execute `pytest --cov=.` post-implementation. Verify quantitative metric thresholds (AUROC, C-index, MAPE).
- ☐ **Check-Off Protocol:** Mark `[x] Completed` on test pass, record logs, and commit with message: `feat(<module>): completed task <id>`. On failure, mark `[REQUIRES_REFINEMENT]`.

4. PHASE 2 – MACHINE LEARNING INTELLIGENCE LAYER (DAYS 5–9)

Model Architecture

TASK ID	COMPONENT MODULE	IMPLEMENTATION SPECIFICATION	TARGET BENCHMARK	STATUS
MOD-01	Clinical NLP	Fine-tune ClinicalBERT/PubMedBERT in <code>models/clinical_nlp/</code> for UMLS/RxNorm NER, clinical note complexity, and diagnosis classification.	AUROC > 0.80 NER F1 > 0.70	Pending
MOD-02	Heterogeneous GNN	PyTorch Geometric GATv2/HeteroConv in <code>models/graph_network/</code> constructing Patient-Disease-Drug graph embeddings with residual connections.	AUROC > 0.78 (Mort) AUROC > 0.72 (Readm)	Pending
MOD-03	Survival Analysis	Longitudinal time-to-event models in <code>models/survival/</code> (Cox PH, DeepSurv, Dynamic-DeepHit) predicting readmission & covenant breach timing.	C-index > 0.70	Pending
MOD-04	Tabular Baselines	XGBoost (with Tweedie/Gamma loss for skewed claims data) and LightGBM hyperparameter optimization in <code>models/tabular/</code> .	Outperform traditional GLM	Pending
MOD-05	Stacking Ensemble	Level-1 Ridge/ElasticNet meta-learner in <code>models/ensemble/</code> combining embeddings via 5-fold time-aware cross-validation.	Lower MAPE than any single model	Pending

5. PHASE 3 – DOMAIN FINANCIAL & ACTUARIAL ENGINES (DAYS 10–12)

Quantitative Engines

TASK ID	COMPONENT MODULE	IMPLEMENTATION SPECIFICATION	TARGET BENCHMARK	STATUS
FIN-01	Actuarial Pricing & Reserving	Health insurance premium pricing engine comparing GLM vs AI; Chain Ladder & Bornhuetter-Ferguson IBNR claim reserving models in <code>financial/insurance/</code> .	MAPE < 15% (Indiv) MAPE < 5% (Cohort)	Pending
FIN-02	Hospital Credit Risk	Dual scorecard model in <code>financial/credit_risk/</code> merging balance sheet metrics (DSCR, Days Cash) with clinical metrics (30-day readmissions, HCAHPS, CMI).	Gini / AR > 0.50 KS Statistic > 0.30	Pending
FIN-03	Pharma Portfolio Analytics	Clinical trial monitor, risk-adjusted Net Present Value (rNPV) Monte Carlo simulator, and Markowitz Mean-Variance optimizer in <code>financial/pharma/</code> .	Sharpe Ratio > 1.0 (Backtest)	Pending

Methodological Note on Time-Aware Validation: To prevent data leakage in longitudinal financial models, all cross-validation split schemes in `MOD-05` and `FIN-01` MUST enforce strict temporal sequencing (train on historical quarters $T_{\{0\}} \dots T_{\{k\}}$, validate on $T_{\{k+1\}}$).

6. PHASE 4 – GAMIFIED HEALTHRISK LAB ENGINE (DAY 13)

Simulation & Gamification

Task ID	Component Module	Implementation Specification	Target Benchmark	Status
SIM-01	Quarterly Cycle Engine	40-quarter (10-year) turn-based state machine in <code>simulation/</code> . Implements 4 game modes (Actuarial, Hospital Credit, Pharma, Master), AI opponent agent, & 1000-pt scoring.	Fully playable cycle execution	[] Pending
SIM-02	Scenario Shock Generator	Exogenous shock simulation engine modeling Pandemics, FDA Black Box Warnings, Medicare Reimbursement Cuts, and Regional Hospital Mergers.	Realistic claim & market response	[] Pending

7. PHASE 5 – GOVERNANCE, EXPLAINABILITY & QA (DAYS 14–15)

Compliance & Robustness

Task ID	Component Module	Implementation Specification	Target Benchmark	Status
EXP-01	Model Explainability	SHAP value decomposition, counterfactual actionability generator, and Partial Dependence Plots (PDP) in <code>explainability/</code> .	EU AI Act & ECOA Compliance	[] Pending
TST-01	Quality Assurance Suite	Comprehensive unit and integration test suite in <code>tests/</code> covering raw data pipelines, ML inferences, and financial engines.	≥ 80% Coverage (Min: 60%)	[] Pending

8. QUALITY ASSURANCE & TESTING MATRIX

Validation Standards

Code Coverage Hard Floor Test coverage measured via <code>pytest --cov</code> must meet or exceed 80% across the repository. Hard disqualification below 60%.	Integration & Environment Validation Execution of <code>docker-compose up --build</code> must initialize all microservices without runtime crashes or missing dependencies.
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9. RESEARCH & DOCUMENTATION DELIVERABLES

Publications & Docs

TASK ID	DELIVERABLE ITEM	REQUIRED CONTENT & STRUCTURAL SPECIFICATIONS	STATUS
DOC-01	README.md	Architecture diagrams (Mermaid format), complete setup commands, reproduction CLI steps, and system benchmarks.	[] Pending
DOC-02	Model Cards	Google Framework Model Cards for Clinical NLP, Heterogeneous GNN, Survival, Ensemble, and Hospital Credit Scorecard.	[] Pending
DOC-03	Technical Report	15–20 page academic/industry paper compiled in LaTeX under <code>reports/</code> with mathematical derivations, prior art, and benchmarks.	[] Pending
DOC-04	API Documentation	Sphinx / MkDocs auto-generated complete code reference committed under <code>docs/</code> .	[] Pending
DOC-05	Simulation Manual	HealthRisk Lab game mechanics, scoring rubrics, scenario parameters, and user operational manual.	[] Pending

10. ZERO-DISQUALIFICATION COMPLIANCE CHECKLIST

Final Sign-Off

☐ **Test Coverage Verification:** `pytest --cov` verified $\geq 60\%$ (Targeting $\geq 80\%$).

☐ **Security Guardrail:** Verified NO API keys, tokens, or credentials are committed in git history or configuration files.

☐ **Container Stability:** Verified `docker-compose up --build` spins up cleanly without runtime errors.

☐ **Data Leakage Audit:** Verified temporal split integrity across all time-series and longitudinal models.

☐ **Artifact Archival:** DVC remotes, MLflow experiments, and compiled PDF report stored under `reports/`.

☐ **Git Release Tagging:** Code committed to `main`; release tag created: `git tag -a v1.0 -m "HealthRisk AI v1.0 Final Submission"`.

Execution Sign-Off & Approval Block

Lead AI Engineer

Quantitative Actuary

Healthcare Data Scientist